

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene

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Under a rigid-strain upper-bound protocol (PBE+D3) with fixed fractional coordinates, all reported coupling magnitudes bound—rather than predict—equilibrium total-energy cross terms until ionic-relaxation benchmarks complete (Table II). Sequential strain and doping scans are standard in covalent molecular networks, but total-energy effects are not necessarily additive. In quasi-hexagonal C₆₀ graphullerene we quantify the omitted cross term with the synergy order parameter $\mathcal{S}$ (meV per atom) [Eq. (1)]. On tetramers ($\sim 5\%$ nominal doping) and fifteen periodic $n \times \text{C}_{60}$ supercells ($n = 1\text{--}8$, B/N/P @ $+3\%$, seed 42 placement; one substituent per C₆₀, ~ 1.67 at.%), the linear strain coefficient α is opposite for N (-303 meV/%) and B ($+57$ meV/%), and the protocol yields $|\mathcal{S}|$ values up to 31.9 meV/atom for P at $n=1$ and $+3\%$ strain, with P showing the largest $|\mathcal{S}|$ despite the smallest $|\alpha|$. Sequential high-throughput screens can introduce systematic errors in the estimation of *relative* stability margins when local dopant distortion couples nonlinearly to external strain. Marcus reorganization energies and nearest-neighbor hopping parameters (Figs. 3 and 4) are reported for transport context but do not affect the non-additivity conclusions.

I. INTRODUCTION

Covalent molecular networks—from graphullerenes to covalent organic frameworks—are often tuned jointly by substitutional chemistry and mechanical load, yet many pipelines still estimate $E_{\text{tot}}(\epsilon, \delta)$ from *separate* doping-only and strain-only calculations [1–3]. Synthetic carbon allotropes with extended π conjugation underpin optoelectronic applications; charge transport in molecular networks is often polaronic and anisotropic [4–7], yet prior work emphasizes band edges, mobility, or hopping rather than a total-energy cross term evaluated at the *identical* exchange-correlation functional level under joint biaxial (ϵ, δ) load. Quasi-hexagonal (qHP) C₆₀ graphullerene [4, 5] provides a covalently linked model network for such coupled tuning, with interfullerene linking that raises hopping couplings relative to vdW C₆₀ monolayers [4]. Ab initio models of pristine qHP networks report intrinsic intramolecular strain from the eight-bond hexagonal connectivity [8], so external biaxial load couples to an already distorted cage baseline. Beyond pristine monolayers, strain and substitution independently reshape band edges [2, 9–11], while related 2D fullerene networks show anisotropic mechanical and tribological response under load [12, 13] and endohedral qHP C₆₀ cage filling reshapes band gaps [14]; none of these reports a periodic total-energy cross term under joint biaxial (ϵ, δ) load, motivating explicit coupled grids rather than an additive superposition of composition and strain effects. Non-additive coupling is documented in other two-dimensional networks [1]; a quantitative *total-energy* synergy order parameter $\mathcal{S}$ [Eq. (1)] has not been reported for substitutional B/N/P qHP C₆₀ tetramers and $n = 1\text{--}8$ supercells under a single PBE+D3 protocol across fifteen periodic (n, δ) data points at $+3\%$ strain (seed 42 placement).

We address three questions on this model network:

(1) how opposite linear strain coefficients α (meV per 1% biaxial strain of the *total* cell energy) emerge for donor vs. acceptor dopants; (2) how large the omitted synergy cross term $|\mathcal{S}|$ (meV/atom) is when strain and substitution are applied jointly; and (3) whether $|\mathcal{S}|$ tracks $|\alpha|$ or instead reflects local structural distortion. Using $\gtrsim 90$ converged CP2K tasks (Table I), we define $\mathcal{S}$ on matched tetramer and periodic PBE+D3 grids (fifteen (n, δ) points at $+3\%$ strain, seed 42 placement) and report opposite donor/acceptor α on the tetramer strain grid, demonstrating that the largest $|\mathcal{S}|$ does not coincide with the smallest tetramer $|\alpha|$ for size-mismatched P. *Local* dopant distortion and *global* biaxial strain interact nonlinearly in total energy: $|\mathcal{S}|$ can exceed what linear α alone would suggest, so *relative* stability margins from $\epsilon = 0$ substitution energies or strain-only elastic increments need not survive a coupled load even if they do not reorder dopants with widely separated E_{sub} values. Marcus λ and hopping observables (Figs. 3 and 4) characterize transport-relevant quantities but are not used to infer $\mathcal{S}$.

II. COMPUTATIONAL METHODS

A. Electronic-structure setup

All calculations use CP2K [15] with PBE [16] and Grimme DFT-D3 (Becke–Johnson damping) [17]; Table I summarizes basis sets, SCF convergence criteria, and the full task inventory. We do not employ the Koopmans-compliant rVV10 approach used in Ref. 4, and instead adopt a uniform PBE+D3 protocol suitable for large-scale parameter scanning, at the cost of absolute accuracy in polaron binding energies.

B. Notation and conventions

Biaxial strain is denoted ϵ (percent); dopant species $\delta \in \{B, N, P\}$. The synergy order parameter is denoted by the calligraphic symbol $\mathcal{S}$, with units of meV per atom [Eq. (1)]. π -projected densities of states are written π -DOS (never bare “pi” in equations).

Rigid-strain protocol. Biaxial strain scales the in-plane lattice vectors of the *pristine* equilibrium cell at fixed fractional coordinates (no ionic relaxation under load). Reported α and $\mathcal{S}$ under this protocol are *upper bounds* at fixed coordinates, not equilibrium thermodynamic quantities. Table II benchmarks four fixed-cell geometry-optimization corners for tetramer $P@+3\%$ to test whether $\mathcal{S}$ retains *sign* after ionic relaxation *within the same tetramer grid* (magnitude ratios are not quantitative until a matched-functional rigid reference exists); it does *not* validate periodic $n=4$ magnitudes, which differ in dopant concentration and boundary conditions. Main-text $|\mathcal{S}|$ values therefore bound—rather than predict—meV/atom-scale non-additivity until Table II converges.

Plane-wave cutoff (size scaling). Archived CP2K inputs use 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table I); comparisons at fixed n always share the same cutoff. Cross- n trends therefore mix a modest plane-wave cutoff change at $n = 6$; we do *not* interpret the reported N $\mathcal{S}$ sign change between $n = 6$ and $n = 8$ as a physical size effect until the matched 400 Ry $n=6$ control in Table I completes.

C. Substitutional doping and grids

Tetramer models ($4 \times C_{60}$, 240 sites) substitute B, N, or P at nominal 5% (12–13 sites from archived coordinates, random seed 42); these dopant configurations are not thermodynamically optimized ground states. A second seed 137 tetramer ensemble (18 single-point energy tasks; Table III) tests whether tetramer α and $\mathcal{S}$ are placement-sensitive at $\sim 5\%$ nominal doping. All periodic supercells propagate the same seed 42 site map (one substituent per C_{60}); quantitative $|\mathcal{S}|$ on the fifteen-point grid is therefore configuration-specific, whereas donor/acceptor sign trends for α are discussed as element-intrinsic qualitative expectations pending Table III. Periodic scaling cells place one substituent per C_{60} (n dopants among $60n$ lattice atoms, ~ 1.67 at.% for all n), contrasting with the tetramer grid ($\sim 5\%$ nominal). Tetramer α , gaps, and tetramer IPR/ J trends (Fig. 4) are therefore not directly interchangeable with periodic $\mathcal{S}(n)$ without a matched-concentration benchmark; all $\mathcal{S}$ claims use the periodic protocol. In total, $\gtrsim 90$ converged neutral tasks support the protocol-audited claims below (Table I); rigid-strain $|\mathcal{S}|$ magnitudes are upper bounds until Tables II and III complete. Charged-state geometry optimizations (UKS, multiplicity 2, EPS-SCF = $10^{-6} E_h$) supply adiabatic ionization potentials and electron affinities ($E_{\text{opt}}^{\pm}$). Eight

vertical single-point energy evaluations compute $E_{\text{vert}}^{\pm}$ at the *neutral*-geometry coordinates with charged UKS settings matched to the relaxed state (EPS-SCF = $10^{-7} E_h$ by default; B $q = -1$: $10^{-6} E_h$), giving $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ following Ref. 4. These Marcus diagnostics appear only in Fig. 3 and do not enter Eq. (1).

D. Energy observables

Linear strain sensitivity $\alpha = dE/d\epsilon$ (meV per 1% biaxial strain, tetramer *total* energy) is obtained from the 24-point grid (Table IV). Positive α means tension *raises* total energy; negative α means tension *lowers* it under the rigid-strain protocol. Substitution energy $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is the total-energy difference per dopant site at fixed strain *without* elemental chemical-potential corrections; it is *not* a thermodynamic standard formation enthalpy ($\sim$ eV-scale tabulated values) and is used only for relative ranking within this protocol. In sequential high-throughput screens one often estimates $E_{\text{tot}}(\epsilon, \delta) \approx E_{\text{sub}}(\epsilon=0) + E_{\text{strain}}(\epsilon)$; the neglected term is precisely $\mathcal{S}(\epsilon, \delta)$ —the absolute error in that additive estimate. For screening workflows under the rigid upper-bound protocol, the magnitude of this additive omission is best gauged against the substitution-energy scale: at $n = 4$ and $+3\%$ strain, P has $|\mathcal{S}| = 23.7$ meV/atom while the per-atom substitution scale is $|E_{\text{sub}}| \times n_{\text{dop}}/n_{\text{atoms}} \approx 0.79$ eV/atom ($|E_{\text{sub}}| \approx 16$ eV per P site, $n_{\text{dop}}=4$ substituents in the $n=4$ cell, Table IV)—i.e. $|\mathcal{S}|/|E_{\text{sub}}|_{\text{per atom}} \sim 3\%$ as an *order-of-magnitude* ratio only, because tetramer and periodic dopant concentrations differ; this ratio is not used as a core nonlinearity test.

Non-additive coupling metric. We quantify strain-doping cross terms with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times C_{60}$. The bracketed term is the *additive* prediction from separate strain-only and doping-only scans; deviations from zero measure non-additive coupling. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} < 0$ indicates cooperative stabilization beyond the two-scan sum, whereas $\mathcal{S} > 0$ indicates cooperative destabilization.

When $\alpha_{\delta} = dE/d\epsilon$ denotes the *total*-cell linear strain coefficient (meV per 1% strain) at fixed n , a first-order estimate of the per-atom cross term is $\mathcal{S}(\epsilon, \delta) \approx (\alpha_{\delta} - \alpha_0) \epsilon / N_{\text{atoms}}$ at small ϵ under the rigid-strain protocol; tetramer and periodic n use different N_{atoms} and dopant concentrations, so this link is not used quantitatively in the periodic audit (Discussion, argument scope).

Periodic-cell $\mathcal{S}$ values are extracted from converged CP2K total energies (Table I); the fifteen-point $+3\%$

strain audit accompanies the open data release. We report $\mathcal{S}$ for total energy only; an analogous gap cross term $\mathcal{S}_{\text{gap}}$ is not computed here—main-text panel (a) shows that gap slopes need not track $|\mathcal{S}|$.

Localization and coupling (Fig. 4). Following Capobianco *et al.* [4], we report the inverse participation ratio (IPR) of the charged-state frontier orbital and the nearest-neighbor electronic coupling J from tetramer dimer fragments. IPR is $N \sum_i |\psi_i|^4 / (\sum_i |\psi_i|^2)^2$ on a Mulliken-projected orbital grid; J is estimated from the ground–first-excited splitting of the anionic dimer at fixed interfullerene geometry (two-state model). These observables are computed at the same PBE+D3 level as the energy scans and are *not* directly comparable in absolute value to the Koopmans-compliant $J \simeq 75$ –81 meV reported for pristine qHP networks [4], but they provide an internally consistent trend under coupled (ϵ, δ) .

Polaron binding. Marcus reorganization energies follow the vertical-adiabatic decomposition [4]: $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$, with separate internal ($\lambda_{\text{loc}}^\pm$) and environmental ($\lambda_{\text{env}}^\pm$) contributions when geometry subsets are relaxed. Charged-state relaxations and vertical single-points are documented in Fig. 3. Charged geometry optimizations supply adiabatic IP/EA; vertical single-points at the neutral geometry yield Marcus $\lambda^\pm$ (Fig. 3).

1. DFT calculator settings

All total-energy and electronic-structure calculations use CP2K [15] with PBE [16] and Grimme DFT-D3 (Becke–Johnson damping) [17]. Unless stated otherwise, biaxial strain scales the in-plane lattice vectors of the pristine equilibrium cell at fixed fractional coordinates (rigid-strain protocol). The plane-wave cutoff is 400 Ry for $n \leq 4$ and 350 Ry for $n \geq 6$ (Table I); the atom-centered basis (DZVP-MOLOPT-SR-GTH) is unchanged across n . Tetramer models ($4 \times \text{C}_{60}$, 240 atoms) use

B/N/P substitution at nominal 5% (12–13 dopant sites, seed 42). Periodic supercells place one substituent per C_{60} unit ($n = 1$ –8), giving n dopants among $60n$ lattice atoms (~ 1.67 at.% for all n).

2. Marcus vertical step and post-processing

Charged-state geometry optimizations supply adiabatic ionization potentials and electron affinities ($E_{\text{opt}}^\pm$). Eight vertical single-point energy evaluations compute $E_{\text{vert}}^\pm$ at the *neutral*-geometry coordinates with the charged UKS settings of the relaxed state, following Ref. 4; $\lambda^\pm = E_{\text{vert}}^\pm - E_{\text{opt}}^\pm$ (Fig. 3). π -projected densities of states use a 0.05 eV Gaussian broadening aligned to the Fermi level (Fig. 1). Inverse participation ratios and hopping integrals J are defined in Sec. II and Fig. 4; numerical values are listed in the Data Availability repository.

E. Validation benchmarks

The full calculation map (converged tasks, main-text usage, and pending controls) is in Table I; ionic-relaxation energies (Table II) and the alternate seed 137 grid (Table III) appear there when complete. Three validation benchmarks—rigid vs. relaxed tetramer $\mathcal{S}_P$ (Table II), alternate dopant seed 137 (Table III), and a matched 400 Ry $n=6$ N control (Table I)—gate interpretation of the N $\mathcal{S}(n)$ sign change cited in Secs. III C and IV. Table II has completed the pristine fixed-cell geometry-optimization corners at $\epsilon=0$ and $+3\%$ ($E = -1363.9551$ and $-1363.9550 E_h$); $\mathcal{S}_{\text{relaxed}}$ and sign retention await the P-doped corners ($P@ \epsilon=0$ running; $P@ +3\%$ queued).

Table IV and Sec. III B report E_{sub} using the substitution-energy definition above (no chemical-potential reference; not a formation enthalpy).

TABLE I. Summary of converged DFT calculations and correspondence to main-text results. The ~ 90 tasks cited in the main text are the converged calculations that directly support quantitative claims; the full archive includes additional preliminary grids (rows marked “not used”).

Calculation set	Model	Converged tasks	Main-text usage	Notes
Early strain probe	$2 \times \text{C}_{60}$	5	not used	superseded by tetramer grids
Single-molecule doping	C_{60}	11	not used	superseded
Early electronics	$2 \times \text{C}_{60}$	24	not used	superseded by tetramer π -DOS
Polaron IPR/ J tetramer dimer	tetramer	4	Fig. 4	pristine/B $\times \epsilon=0, +3\%$
Tetramer strain energies	tetramer	24	Table IV, α	rigid strain
Optimal-design grid	tetramer	20	not used	archive only
Tetramer π -DOS/gaps	tetramer	12	Fig. 1; main panels	
Geometry references	tetramer relax	6	charged-state inputs	reference geometries
Charged polarons	tetramer $q = \pm 1$	12 geometry opt.	Fig. 3	Marcus $\lambda^\pm$ and adiabatic IP/EA
Size scaling	$n \times \text{C}_{60}$	40 (+1 pending)	main $\mathcal{S}(n)$	400 Ry ($n \leq 4$), 350 Ry ($n \geq 6$); cutoff400 contr
$n=6$ cutoff control	$6 \times \text{C}_{60}$ N @+3%	pending	N sign benchmark	400 Ry matched control (pending)
Ionic relaxation (tetramer)	tetramer	2/4 geometry opt.	Table II	fixed-cell; pristine $\epsilon=0, +3\%$ converged

III. RESULTS

We organize the results around total-energy non-additivity under the rigid-strain protocol (Sec. II B):

Sec. III A summarizes band-edge context; Sec. III B re-

TABLE II. Ionic-relaxation benchmark for tetramer $P@+3\%$ (four fixed-cell geometry-optimization corners, same biaxial cell). This benchmark is *sign-qualitative only*: the rigid reference uses legacy PBE without D3, whereas relaxed corners use PBE+D3, so $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}|$ is *not* a quantitative retention ratio until a matched-functional rigid grid is recomputed. We test whether $\mathcal{S}$ retains *sign within the tetramer grid only*; it does *not* calibrate periodic main-text $\mathcal{S}(n)$, which uses different dopant concentration and boundary conditions. Rigid reference energies: archived fixed-coordinate tetramer single-point grid (legacy PBE, no D3). Relaxed energies: PBE+D3 fixed-cell geometry optimization (400 Ry) from archived inputs in the open repository. $\mathcal{S}$ follows Eq. (1) with $N=240$ atoms.

Quantity	Rigid (legacy tetramer)	Relaxed (geometry opt.)
$E(\text{pristine}, \epsilon=0) (E_h)$	-1363.5512	-1363.9551
$E(\text{pristine}, \epsilon=3\%) (E_h)$	-1363.5522	-1363.9550
$E(P, \epsilon=0) (E_h)$	-1370.4965	pending
$E(P, \epsilon=3\%) (E_h)$	-1370.4890	pending
$\mathcal{S}$ (meV/atom)	+0.96	pending
Sign of $\mathcal{S}$ vs. rigid	pending (2/4 geometry opt.; $P@+3\%$)	pending

TABLE III. Alternate random-seed dopant placement (*tetramer only*, rigid strain, PBE+D3). Seed 42 defines all main-text tetramer α and the periodic supercell site map (one substituent per C_{60}). Seed 137 repeats the full B/N/P strain grid ($\epsilon \in \{-5, -2.5, 0, +2.5, +3, +5\}\%$; 18 single-point tasks) to test whether tetramer α and $\mathcal{S}$ at $+3\%$ are placement-sensitive at $\sim 5\%$ nominal doping. Periodic $\mathcal{S}(n)$ is *not* tabulated here because concentration and boundary conditions differ; main-text periodic magnitudes remain seed 42-specific until alternate periodic placements are computed.

Dopant	Seed	α (meV/%)	$\mathcal{S}_{+3\%}^{\text{tet}}$ (meV/atom)	Status
B	42	+56.9	$\sim +0.7$	verified (Table S1)
N	42	-303.2	~ -3.8	verified
P	42	-19.1	~ -0.2	verified
	137	pending	pending	6 inp / pending
	137	pending	pending	6 inp / pending
	137	pending	pending	6 inp / pending

ports linear strain coefficients, E_{sub} , and geometric strain response (Table V); Sec. III C gives $\mathcal{S}(n)$ at $+3\%$ strain across dopants and supercell sizes (Table I); Sec. III D reports Marcus λ and IPR/ J after the $\mathcal{S}$ audit.

A. Electronic structure under strain and doping

Figure 2(a,b) summarizes tetramer HOMO-LUMO gaps (archived tetramer electronic-structure set) and representative π -DOS (B @ $\epsilon=0$; color scheme follows Ref. 4). At $\epsilon=0$, B is near a closed π gap ($E_g \approx 0.03$ eV), N doping pins the Fermi level inside the conduction band ($E_g \approx -0.14$ eV), and P remains gapped ($E_g \approx 0.05$ eV); tensile strain narrows the pristine gap (~ 23 meV/% on the three available strain points in the dataset). Donor vs. acceptor ordering of the $\epsilon=0$ gaps tracks the opposite α signs in Sec. III B, but band-edge slopes are not used to infer $\mathcal{S}$.

B. Mechanical strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%).

TABLE IV. Total energies, linear strain sensitivities, and substitution energies for tetramer models (PBE+D3, rigid-strain protocol). $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon=0$ with no chemical-potential reference (not a standard formation enthalpy). n_{dopants} is counted from archived input coordinates (nominal 5% target; B: 13, N: 12, P: 12).

System	n_{dop}	$E(\epsilon=0) (E_h)$	$E(\epsilon=+5\%) (E_h)$	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	-1363.5512	-1363.5509	-1.0	—
B-doped	13	-1328.4900	-1328.4090	+56.9	+73.4
N-doped	12	-1414.3812	-1414.4053	-303.2	-115.3
P-doped	12	-1370.4965	-1370.4888	-19.1	-15.8

On the seed 42 tetramer grid (Table IV), substitution opens an order-of-magnitude spread in α :

N (-303.2 meV/%, tension stabilizes the doped cell), B ($+56.9$ meV/%, tension destabilizes), and P

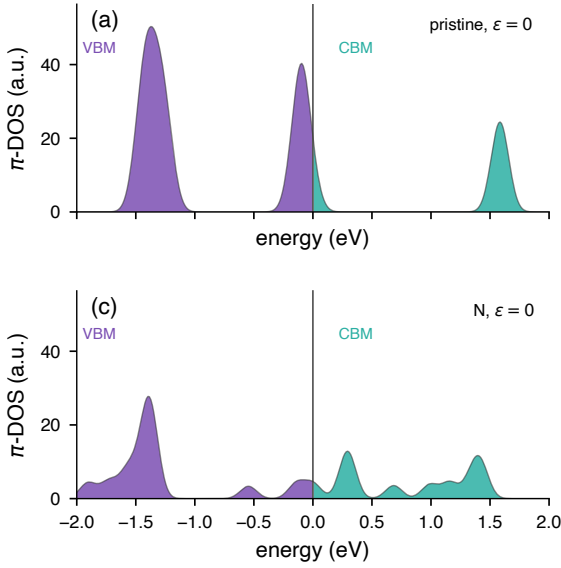


FIG. 1. π -projected density of states (π -DOS) for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3). Panels (a)–(d) share a common vertical scale; energy is relative to the Fermi level at 0 eV (Capobianco color scheme [4]).

TABLE V. Local dopant–cage structure from archived tetramer XYZ (mean nearest C distance $\bar{d}$ in Å; standard deviation $\sigma(\bar{d})$ across dopant sites; covalent-radius excess vs. C in pm). Strain-induced change $\Delta\bar{d} = \bar{d}(\epsilon = +3\%) - \bar{d}(\epsilon = 0)$ summarizes how tensile load couples to pre-existing distortion (Discussion mechanistic synthesis (ii),(vi)). Values are computed from archived tetramer XYZ coordinates (seed 42) in the open data release; Mayer bond order and Bader charges are not included.

Dopant	$\bar{d}(\epsilon=0)$ (Å)	$\bar{d}(\epsilon = +3\%)$ (Å)	$\Delta\bar{d}$ (Å)	$\sigma(\bar{d})$ @ +3% (Å)	Δr_{cov} (pm)
B	1.384	1.393	+0.0085	0.019	+7
N	1.393	1.402	+0.0087	0.024	−6
P	1.384	1.385	+0.00003	0.0004	+30

Table V links this mechanical spread to local geometry: P shows negligible strain-induced change in mean nearest C distance ($\Delta\bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B at +3% load), foreshadowing the α – $\mathcal{S}$ decoupling quantified in Sec. III C (Discussion mechanistic synthesis).

C. Non-additive strain–doping coupling

Figure 2(d) tracks protocol $\mathcal{S}(n)$ (rigid-strain upper bounds) with one dopant per C_{60} at +3% biaxial strain. At $n = 4$ under the seed 42 periodic site map (alternate placements not yet computed), $\mathcal{S} = -6.2$ meV/atom (B), +3.0 meV/atom (N), and −23.7 meV/atom (P); P shows the largest magnitude despite the smallest $|\alpha|$ on the seed 42 tetramer grid (Table IV). Across fifteen seed 42 periodic points ($n \in \{1, 2, 4, 6, 8\}$, B/N/P), $|\mathcal{S}|$ reaches 31.9 meV/atom for P at $n = 1$ and remains $\mathcal{O}(10)$ meV/atom at larger n for B and P, whereas N is

smaller in magnitude at $n = 4$ but comparable at $n = 1$. For N, $\mathcal{S}$ changes sign between $n = 6$ (+2.4 meV/atom) and $n = 8$ (−2.3 meV/atom) at the production 350 Ry cutoff; this reversal is *not* used in the main conclusions until the matched 400 Ry $n=6$ control in Table I and Tables II and III complete (Sec. II E). Core size-scaling claims focus on $n \leq 4$ at matched 400 Ry cutoffs; B and P show monotonic $|\mathcal{S}|$ decrease from $n = 1$ toward $n = 4$, whereas $n \geq 6$ trends are reported for completeness only and mix the 350 Ry production cutoff. We report provisional $\mathcal{S}(n) \approx \mathcal{S}_\infty + A/n$ fits in the open data release but refrain from drawing conclusions about the asymptotic $\mathcal{S}_\infty$ value until cutoffs are matched across n . All periodic $\mathcal{S}$ values quoted in this subsection use the seed 42 dopant-site map and are tabulated in Table I (fifteen $n \times$ dopant points at +3% strain). Negative $\mathcal{S}$ (B, P at $n = 4$) means coupled tension *stabilizes* the doped cell relative to the additive estimate; positive $\mathcal{S}$ (N at $n = 4$) means sequential screens *overestimate* stability under load.

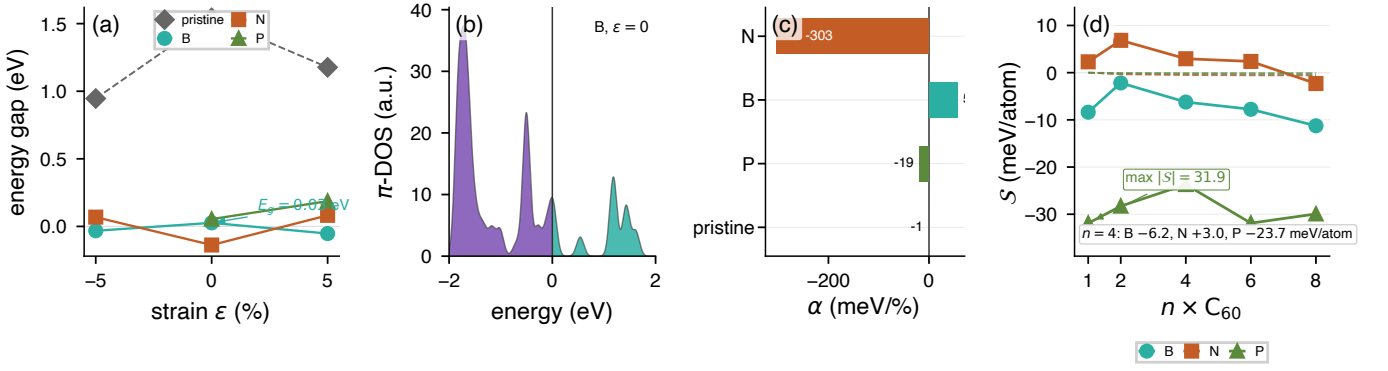


FIG. 2. Coupled (ϵ, δ) response in doped qHP C₆₀ (PBE+D3). (a) HOMO-LUMO gap vs. strain. (b) π -DOS for B @ $\epsilon=0$ (color scheme follows Ref. 4). (c) Mechanical α (Table IV). (d) Seed 42 protocol $S(n)$ at +3% (max $|S|=31.9$ meV/atom for P @ $n=1$; at $n=4$: -6.2 , $+3.0$, -23.7 meV/atom for B/N/P); Marcus λ and IPR/ J in Figs. 3 and 4.

D. Polaron transport diagnostics

Figure 3 reports Marcus reorganization energies from twelve charged-state geometry optimizations and

eight vertical single-points at the neutral geometry (Sec. IID 2).

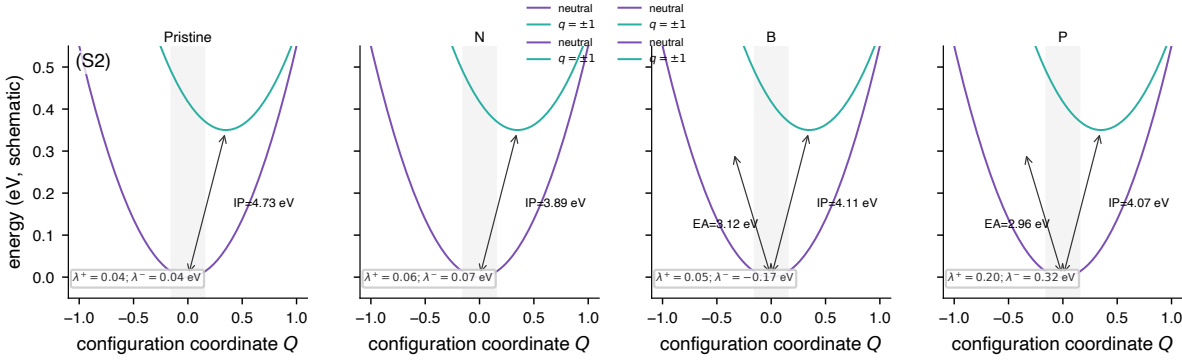


FIG. 3. Marcus reorganization energies (12/12 charged-state geometry optimizations; eight vertical single-points at the neutral geometry). Pristine $\lambda^\pm=0.041/0.042$ eV; N $\lambda^+=0.058$, $\lambda^-=0.065$ eV; B $\lambda^+=0.051$ eV; P $\lambda^+=0.200$, $\lambda^-=0.315$ eV. The electron-channel reorganization energy λ^- for B doping is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

On the tetramer dimer grid, B substitution at $\epsilon=0$ delocalizes the charged-state orbital (IPR $75.0 \rightarrow 24.6$) while raising J from 26.6 to 46.4 meV; coupled B @ +3% gives $J = 37.2$ meV and IPR 23.1. The criterion $J > \lambda/2$ for polaron hopping dominance [4] is *not* met at PBE+D3 ($J < \lambda/2$), so a polaron transition is not confirmed on this grid.

IV. DISCUSSION

Context. Relative to strain-only, doping-only, mobility, or mechanical studies in qHP and related fullerene networks [1, 2, 4, 7–14, 18, 19], the distinction here is

a periodic *total-energy* audit of joint biaxial (ϵ, δ) load [Eq. (1)] on a rigid-strain upper-bound protocol at the identical PBE+D3 functional level. In other 2D hosts, substitutional defects can raise or lower formation energies under biaxial strain depending on dopant–host size mismatch [19]; isomer-engineered C₇₀ network designs similarly tune band edges through topology choice rather than joint substitutional–mechanical load [20]; we instead quantify the omitted additive cross term S in total energy. Quasi-harmonic thermal-expansion studies of qHP monolayers report anisotropic positive and negative lattice response tied to interfullerene bond topology [10, 18], whereas our audit holds fractional coordinates fixed at zero-temperature total energy (upper

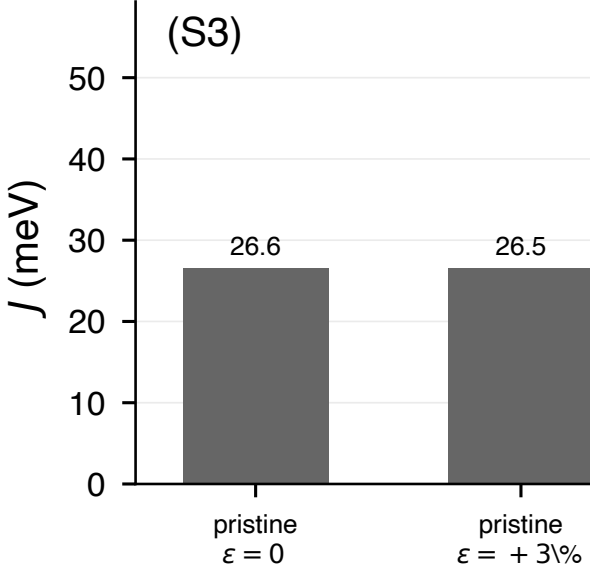


FIG. 4. Nearest-neighbor electronic coupling J on the tetramer dimer grid (pristine and B-doped cells at $\epsilon=0$ and $+3\%$; PBE+D3).

bound pending Table II). Strain-engineering studies on smaller fullerene graphullerenes instead emphasize magnetic or half-metallic transitions [21], which is orthogonal to the neutral total-energy audit pursued here. Band-edge, spin-resolved, and hopping metrics (Fig. 4) contextualize mechanism but are not used to infer $\mathcal{S}$.

Mechanistic synthesis. (i) **Opposite α for N vs. B (seed 42).** N introduces excess π electrons that strengthen bonding under tension (negative α), whereas B creates electron deficiency and antibonding π^* character that rises with lattice expansion (positive α). Quantitatively, the $\epsilon=0$ gap ordering ($E_g^N < E_g^B < E_g^P$ within PBE; Sec. III A) and dopant-resolved π -DOS in Fig. 2(b) and Fig. 1 align with donor vs. acceptor π -counting, while total-energy α spans ~ 360 meV/% between N and B on the seed 42 grid (Table IV). The opposite N vs. B sign is reported for this fixed placement across $\epsilon \in \{-5, \dots, +5\}\%$ (Table IV); Table III repeats the full tetramer strain grid for seed 137 to test slope sensitivity at $\sim 5\%$ nominal doping without mixing periodic boundary conditions. Mayer bond order and Bader charge partitioning along the strain path are not yet computed (Limitations); periodic $\mathcal{S}(n)$ magnitudes are likewise not averaged over alternate supercell placements until such audits exist. (ii) **Large $|\mathcal{S}|$ for P despite small $|\alpha|$.** P carries the largest covalent-radius excess ($+30$ pm vs. C; Table V) and pre-existing cage distortion at $\epsilon=0$, yet shows the smallest strain-induced change in mean nearest C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å vs. $\sim 8.5 \times 10^{-3}$ Å for B/N) and the smallest site-to-site spread at $+3\%$ ($\sigma(\bar{d}) \approx 4 \times 10^{-4}$ Å vs. $\sim 2 \times 10^{-2}$ Å for B/N; Table V). Tensile load therefore couples to *pre-existing* cage distortion from size mismatch rather than

to a uniform elastic response, yielding a cross term invisible in separate scans but dominant when both strain and substitution are applied simultaneously—consistent with P’s intermediate E_{sub} yet the largest $|\mathcal{S}|$ at $n=4$ (23.7 meV/atom; Table I) despite weak tetramer $|\alpha|$. Sec. III B presents this Table 37.2 geometric decoupling before the $n=4$ $\mathcal{S}$ ranking in Sec. III C: tetramer $|\alpha|$ ranks P as weakly strain-sensitive, whereas the coupled periodic audit exposes the largest cross term. (iii) **N $\mathcal{S}$ sign change with n .** With one substituent per C_{60} , electrostatic fields dilute at large n ; any N sign flip between $n=6$ and $n=8$ is confounded with the 400→350 Ry cutoff step and remains uninterpreted (Sec. II E). Together, (i)–(iii) suggest a taxonomy for covalent molecular networks: *electronic* donor/acceptor control of α , *structural* local-distortion coupling to external strain (P), and *finite-size* electrostatic dilution (N at large n). (iv) **Geometric vs. electronic nonlinearity.** For P, Table V shows that rigid tensile load barely moves the mean dopant–C distance ($\Delta \bar{d} \approx 3 \times 10^{-5}$ Å) yet $\mathcal{S}$ remains the largest at $n=4$ (23.7 meV/atom), whereas B/N exhibit $\sim 10^2 \times$ larger $\Delta \bar{d}$ at $+3\%$ but smaller $|\mathcal{S}|$ (6.2 and 3.0 meV/atom). At $n=4$ the coupled correction for P exceeds the rigid per-atom strain increment from tetramer α by two orders of magnitude ($|\mathcal{S}| \gg \alpha \times 3\%/N_{\text{atoms}}$; Methods energy observables), even though $|\alpha|$ ranks P as the weakest strain responder on the seed 42 tetramer grid. (v) **Argument scope (concentration and boundary conditions).** We do *not* validate $\mathcal{S}$ by extrapolating tetramer α to periodic $n=4$: the $\sim 5\%$ tetramer vs. ~ 1.67 at.% periodic concentrations and tetramer vs. periodic boundary conditions are unmatched, so a first-order $(\alpha_\delta - \alpha_0)\epsilon/N_{\text{atoms}}$ estimate is excluded from the argument. Non-additivity is assessed only through Eq. (1) on matched four-corner grids at common (ϵ, δ) ; P in particular requires explicit coupled evaluation rather than sequential α screening. (vi) **Decomposition outlook.** The Table V decoupling indicates that the cross term is not reducible to uniform bond stretching alone; electronic rehybridization and pre-stored cage strain likely contribute, but their relative weights are not partitioned here without Mayer/Bader analysis along the strain path (Limitations).

(vii) **Significance versus DFT uncertainty.** $\mathcal{S}$ is a difference of total energies computed with the *same* functional, basis, and cell for all four corners of Eq. (1), so systematic errors largely cancel in sign and relative magnitude. At $n=4$ and $+3\%$ strain, $|\mathcal{S}| = 23.7$ meV/atom for P is $\sim 3\%$ of the per-atom E_{sub} scale (Table IV) but still $\gg$ rigid per-atom strain increments from α . Large absolute uncertainties in PBE E_{sub} therefore do not invalidate the conclusion that sequential scans omit a meV/atom-scale cross term.

Design implications. At $n=4$ and $+3\%$ strain the coupled correction is $|\mathcal{S}| = 23.7$ meV/atom for P vs. 6.2 meV/atom for B—sequential *additive* estimates differ by approximately a factor of four in the load-dependent *margin* even though widely separated E_{sub} ranks at $\epsilon=0$

(e.g., B vs. P; Table IV) are unchanged. Dopants with larger covalent-radius excess (P: +30 pm vs. B: +7 pm; Table V) show the largest protocol $|\mathcal{S}|$ under fixed coordinates; *size-mismatched* substituents may warrant coupled (ϵ, δ) evaluation rather than one-parameter pipelines once Table II confirms retention after relaxation. The fifteen-point periodic grid tabulates protocol $|\mathcal{S}|$ at each (n, δ) as an audit of additive omission under the rigid protocol (Table I), not as quantitative design targets. At +3% strain, P carries the largest additive-approximation error in protocol $\mathcal{S}$ ($|\mathcal{S}| = 23.7\text{--}31.9$ meV/atom from $n=4$ to $n=1$ under fixed coordinates); qualitative co-design should treat P under tension as a *coupled* stability problem pending Table II, not $|E_{\text{sub}}|$ at $\epsilon = 0$ alone. Where $\mathcal{S} < 0$ (B and P at $n=4$), coupled tension can stabilize doped cells beyond additive predictions; where $\mathcal{S} > 0$ (N at $n=4$), sequential screens overestimate stability under load. For chemistries with comparable $|E_{\text{sub}}|$ or in multi-dopant competition, omitting $\mathcal{S}$ can shift margins by several percent of the per-atom substitution scale—much larger than the rigid strain increment from α —even when the global dopant rank at $\epsilon=0$ is unchanged. Gap closing [Fig. 2(a)] does not track $|\mathcal{S}|$; we do not compute $\mathcal{S}_{\text{gap}}$, so photonic or transport targets cannot be inferred from total-energy $\mathcal{S}$ alone. For B at $n = 4$, both negative $\mathcal{S}$ and positive α imply that a strain-only elastic estimate would predict tension *raises* energy while the coupled scan *lowers* it relative to the additive model—a direct sign conflict that sequential pipelines cannot resolve without evaluating $\mathcal{S}$. For N at $n=4$, negative tetramer α and positive $\mathcal{S}$ present a second sign conflict: tension stabilizes the doped tetramer in isolation, yet the periodic coupled audit reports a positive cross term that sequential additive models fail to capture at ~ 1.67 at.% doping. N $\mathcal{S}$ vs. n is reported for completeness but excluded from design rules until Tables I–III controls converge.

Validation protocol (Tables I–III). Three validation benchmarks characterize protocol limits without altering the seed 42 periodic magnitudes quoted above. **Table II** runs fixed-cell geometry optimization on the four Eq. (1) corners for tetramer $P@ +3\%$ only; because rigid references use legacy PBE without D3 whereas relaxed corners use PBE+D3, the acceptance criterion is *sign retention* of $\mathcal{S}$ after ionic relaxation within that tetramer grid—not a quantitative $|\mathcal{S}_{\text{relaxed}}|/|\mathcal{S}_{\text{rigid}}|$ ratio until a matched-functional rigid grid is recomputed. It does *not* rescale periodic $\mathcal{S}(n)$ because concentration and boundary conditions differ. **Table III** tests tetramer α and tetramer $\mathcal{S}_{+3\%}$ for seed 137 vs. seed 42; qualitative claims about opposite N/B α signs are configuration-specific until this grid converges. **Table I** (cutoff400 row) isolates the 400→350 Ry confound for $6 \times C_{60}$ N @ +3% before any $n \geq 6$ design statement is restored.

Polaron transport. The Marcus workflow in Figs. 3 and 4 parallels Ref. 4—charged-state geometry optimizations ($E_{\text{opt}}^{\pm}$) plus vertical single-points at the neutral geometry ($E_{\text{vert}}^{\pm}$)—at the uniform PBE+D3 level used for $\mathcal{S}$. Recent work shows that charged-supercell and

neutral-defect semilocal routes to polaron self-interaction correction are formally equivalent when the polaronic structure is held fixed [22]; our Marcus λ protocol instead follows the charged vertical/adiabatic split of Ref. 4 without piecewise-linear or γ/μ_{DFT} tuning, and remains orthogonal to the $\mathcal{S}$ audit. Where that study linked hopping J and frontier-orbital localization to mobilities in pristine qHP networks, we retain the same observables for comparability (Fig. 4) but do *not* extend their mobility-centric ranking to neutral total-energy margins. Endohedral band-gap tuning [14] and substitutional $\mathcal{S}$ address a second orthogonal knob (cage filling vs. cage substitution) and are not interchangeable for sequential-screen corrections.

Limitations and outlook. Three protocol limits are actively benchmarked (Table I; Sec. II E): rigid strain (Table II), tetramer alternate-seed placement (Table III), and mixed 400/350 Ry cutoffs across n (Table I cutoff400 control). All periodic $\mathcal{S}(n)$ use a single seed 42 dopant configuration; magnitudes are not averaged over placement ensembles, and periodic alternate-site audits are not yet computed. Additional limits: tetramer vs. periodic concentration mismatch; PBE+D3 without Koopmans/rVV10; $\mathcal{S}_{\text{gap}}$ not quantified; Mayer bond-order and Bader charge partitioning along the strain path are not computed (Table V reports geometric metrics only); the electron-channel reorganization energy λ^- for B doping is unphysical under the fixed neutral-geometry vertical protocol and is excluded from rate estimates (as further discussed in the Limitations subsection); mobility tensors are not computed—polaron transport anisotropy in related fullerene networks is treated only comparatively [4, 6] (Figs. 3 and 4).

V. CONCLUSIONS

Sequential strain-doping workflows in covalent molecular networks can omit a meV/atom-scale total-energy cross term when local dopant distortion couples nonlinearly to external biaxial load. In qHP C_{60} graphullerene we answered the three Introduction questions under a uniform PBE+D3 rigid-strain protocol: **(1)** opposite donor/acceptor α (N vs. B; Table IV); **(2)** meV/atom-scale omitted cross terms on a fifteen-point periodic grid (seed 42; upper bounds at fixed coordinates; Table I); **(3)** $|\mathcal{S}|$ decoupled from tetramer $|\alpha|$ for size-mismatched P via pre-existing cage distortion (Table V). Ionic-relaxation, alternate-seed, and matched-cutoff controls (Tables I–III) will test whether these audits remain relevant after relaxation and placement averaging. Coupled (ϵ, δ) screening is advisable for size-mismatched substituents and close substitution-energy competitors, complementing strain-only mechanical [12], derivative-network [11], and endohedral band-edge [14] axes. Future work extends seed 137 to periodic cells, adds Mayer/Bader strain-path partitioning, and explores polaron-aware semilocal corrections beyond the Marcus

vertical protocol.

DATA AVAILABILITY

The data that support the findings of this study are openly available at <https://github.com/chenxingqiang/graphullerene-strain-engineering>, including converged computational inputs and outputs, tabulated synergy and polaron analyses, and materials to reproduce all figures and tables in the main text. Pending ionic-relaxation, alternate-seed, and matched-cutoff validation tasks are documented in Tables I–III. A tagged release matching the manuscript revision will be issued upon acceptance.

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